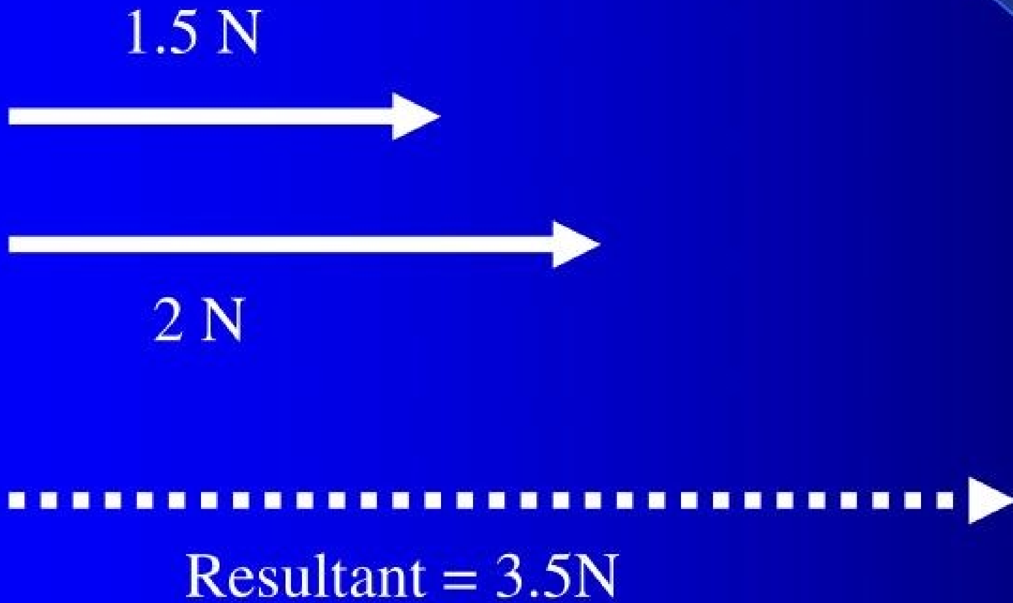
Chapter 4 Vectors

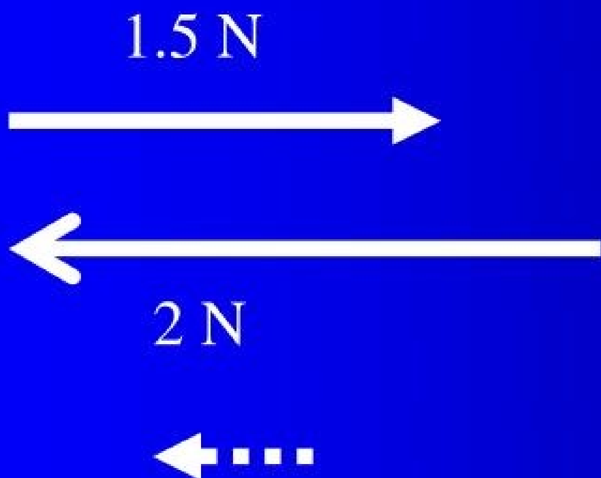
- **Vector** quantities have both **magnitude** and **direction**
- **Scalar** quantities only have **magnitude**
- Velocity, acceleration, displacement, and Force are vector quantities

Vector Addition



The resultant is the sum of the vectors

Vector Addition

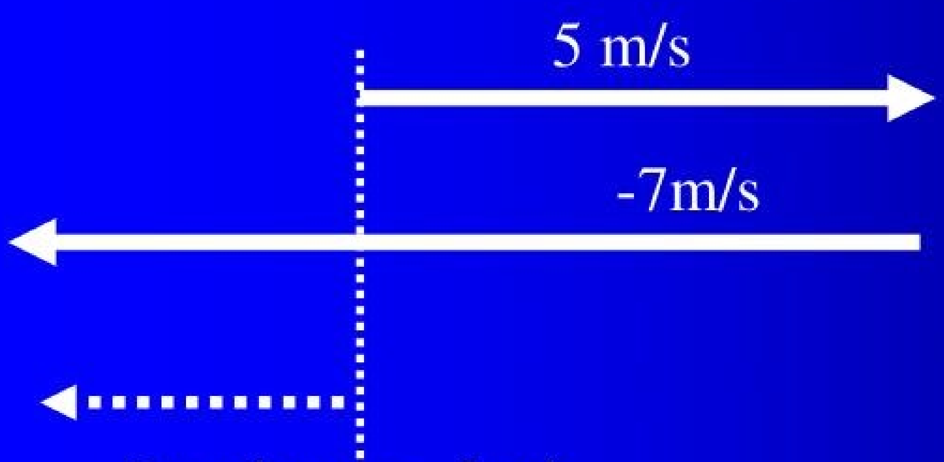


Resultant = - 0.5N

The resultant is the sum of the vectors

Vector addition

- When doing vector addition graphically, the vectors must be drawn to scale and in the correct direction.
- The resultant is found by adding them “Tip to Tail”



Resultant = -2m/s

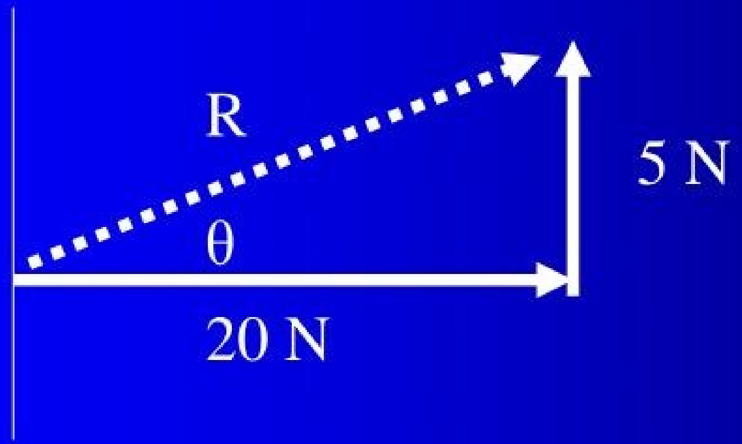
The resultant is drawn from the initial reference point to the tip of the last vector

Vector addition at Right Angles to each other

- **You can never add 2 vectors “Tail to Tail”**

One of the vectors needs to be moved to the “tip” of the other
So that it is “Tip to Tail”

Vector addition at Right Angles to each other



$c^2 = a^2 + b^2$

$C^2 = (20\ \mathrm{N})^2 + (5\mathrm{N})^2$

Resultant = 20.6 N

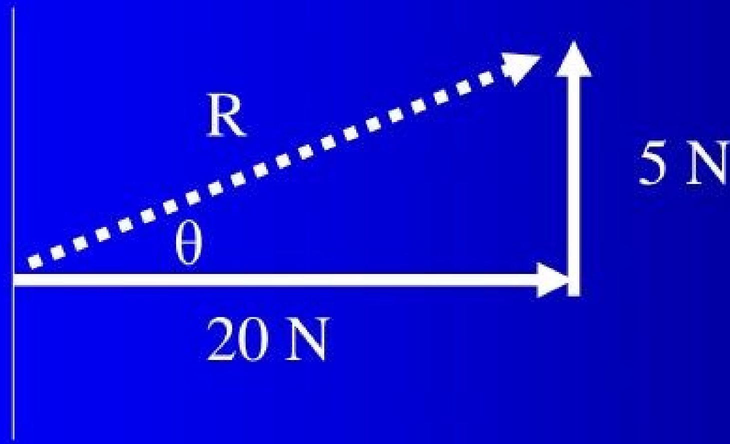
Trig. Functions

- $\text{Sin}\ \theta = \dfrac{\text{Opp.}}{\text{Hypotenuse}}$

  $\text{Cos}\ \theta = \dfrac{\text{Adj}}{\text{Hypotenuse}}$

  $\text{Tan}\ \theta = \dfrac{\text{Opp}}{\text{Adj.}}$

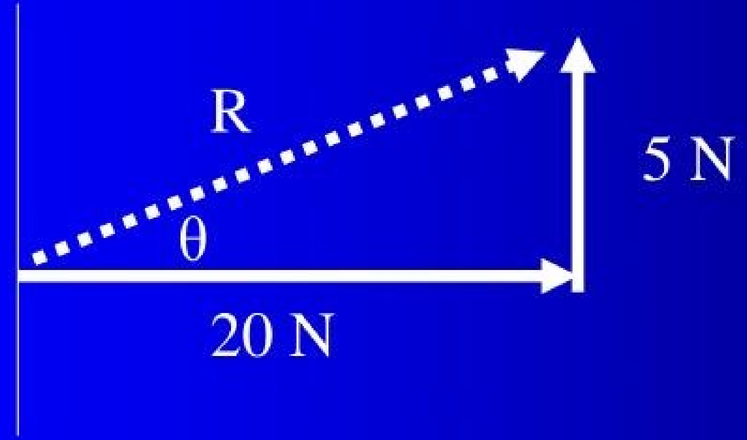
Vector addition at Right Angles to each other



$\mathrm{Tan}^{-1}\,\theta = \frac{\mathrm{opp}}{\mathrm{adj}}$ $\mathrm{Tan}^{-1}\,\theta = \frac{5\mathrm{N}}{20\mathrm{N}}$ $\theta = 14°$

Vector addition at Right Angles to each other

The final answer is 20.6 N @ 14°



Vectors in all quadrants

A plane flies due west at 125 m/s. A wind blows due North at 15 m/s. What is the plane's velocity?

One of the vectors needs to be moved so that it is "tip to tail"

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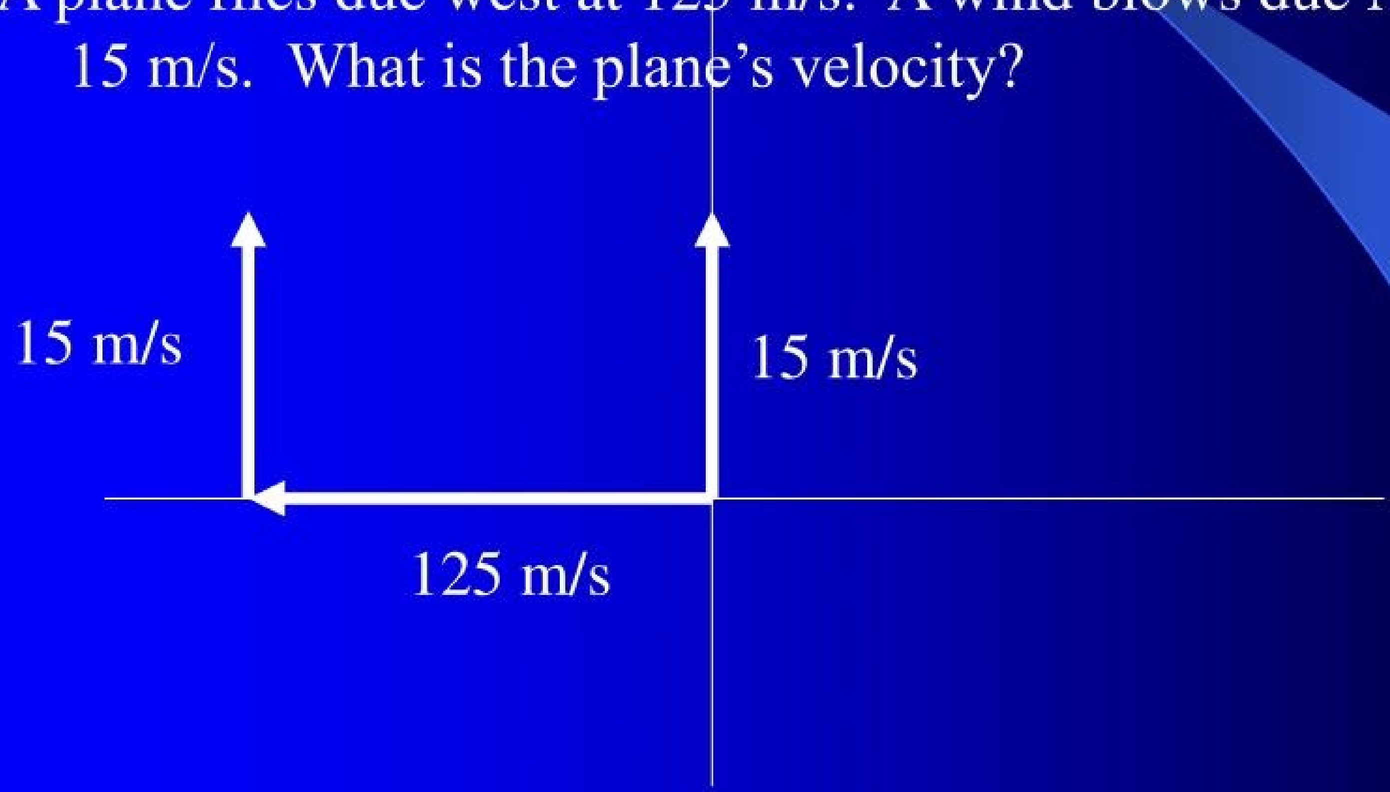

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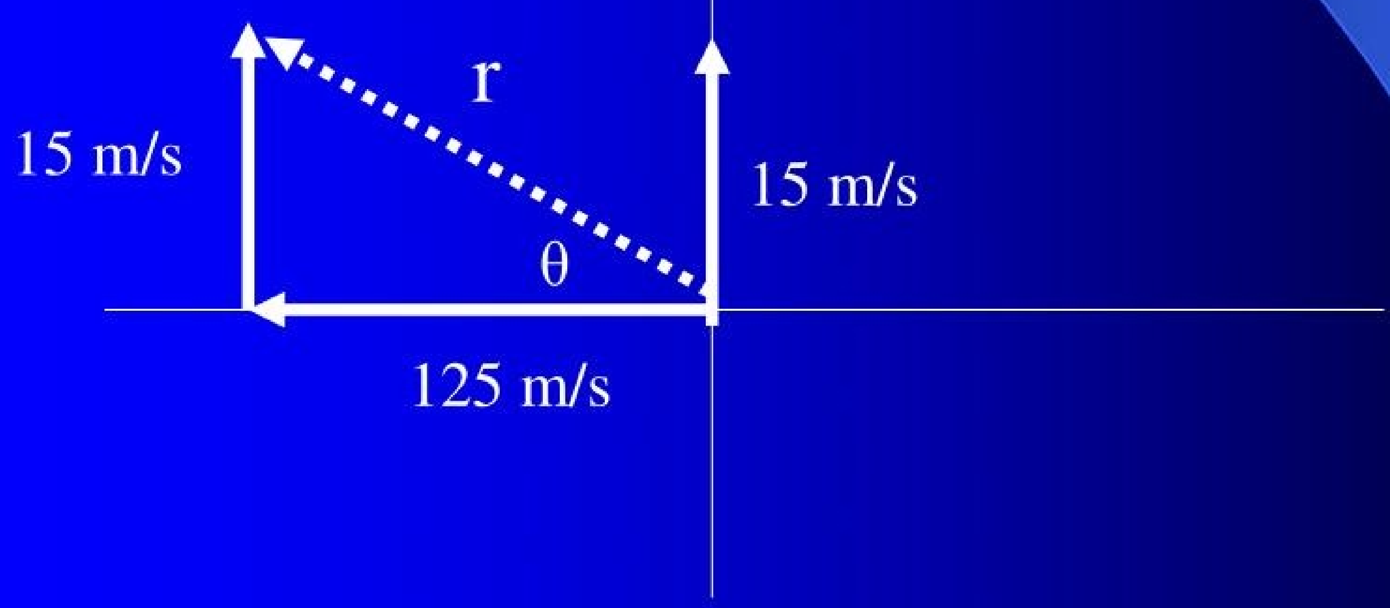

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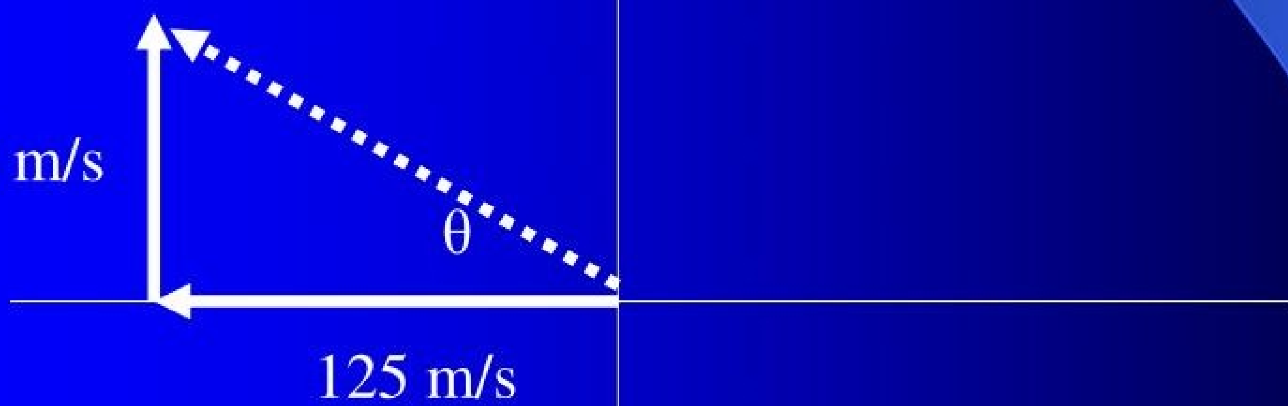

$c^2 = (15 m/s)^2 + (125 m/s)^2$

Resultant = 125.9 m/s

Vectors in all quadrants

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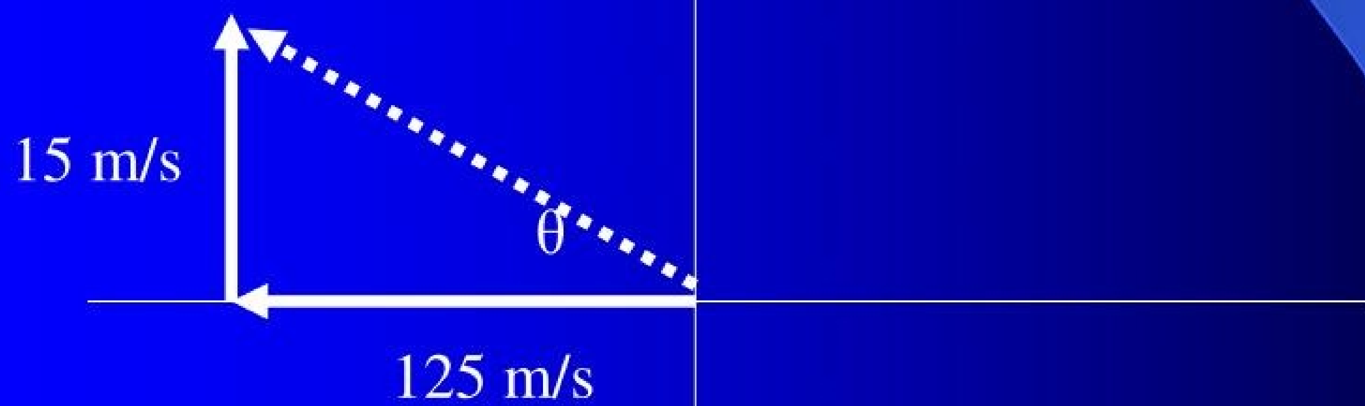


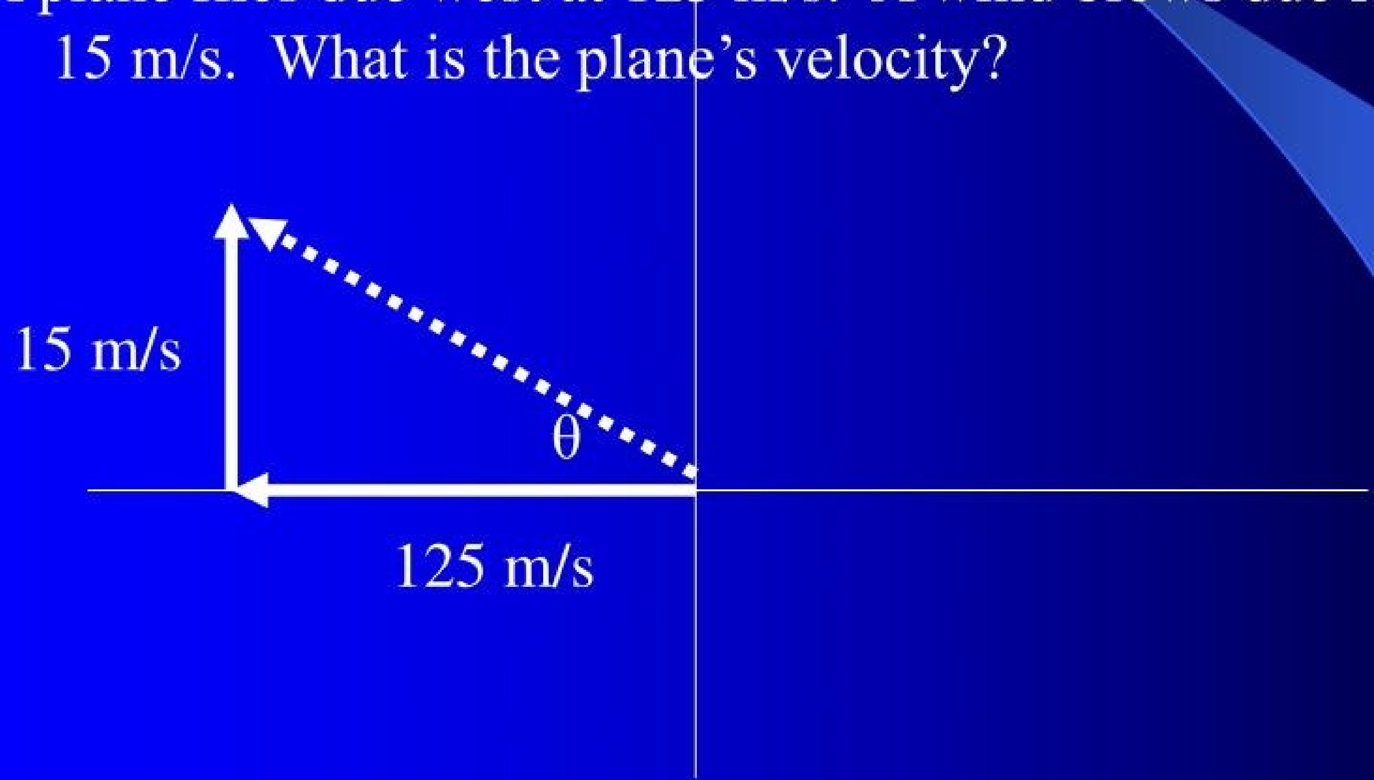
$$\mathrm{Tan}^{-1}\,\theta = \frac{15\,\mathrm{m/s}}{125\,\mathrm{m/s}}$$

$\theta =$ 6.8° N of W

Or 173.2°

Vectors in all quadrants

A plane flies due west at 125 m/s. A wind blows due North at 15 m/s. What is the plane's velocity?



The final answer is 125.9 m/s @ 6.8 N of W

Components of Vectors

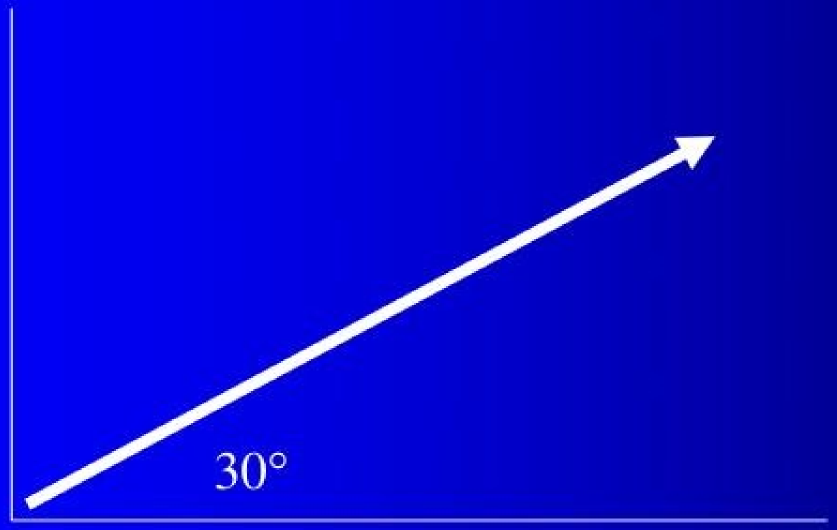
- In the previous problems, you were to find the resultant of 2 or more vectors.
- In a component problem, the resultant is given along with a direction and you need to determine what 2 vectors would give that resultant.
- The 2 vectors are always on the X and Y axis

Components of Vectors

- The following symbols are used to represent various components:
- Horizontal Force F_h
- Vertical Force F_v
- Horizontal velocity v_h
- Vertical velocity v_v
- Any vector quantity can have 2 components

Component Vectors

A rope makes a 30 ° angle with the ground.
If you pull with a force of 58 N, what is the vertical and horizontal components of the force?



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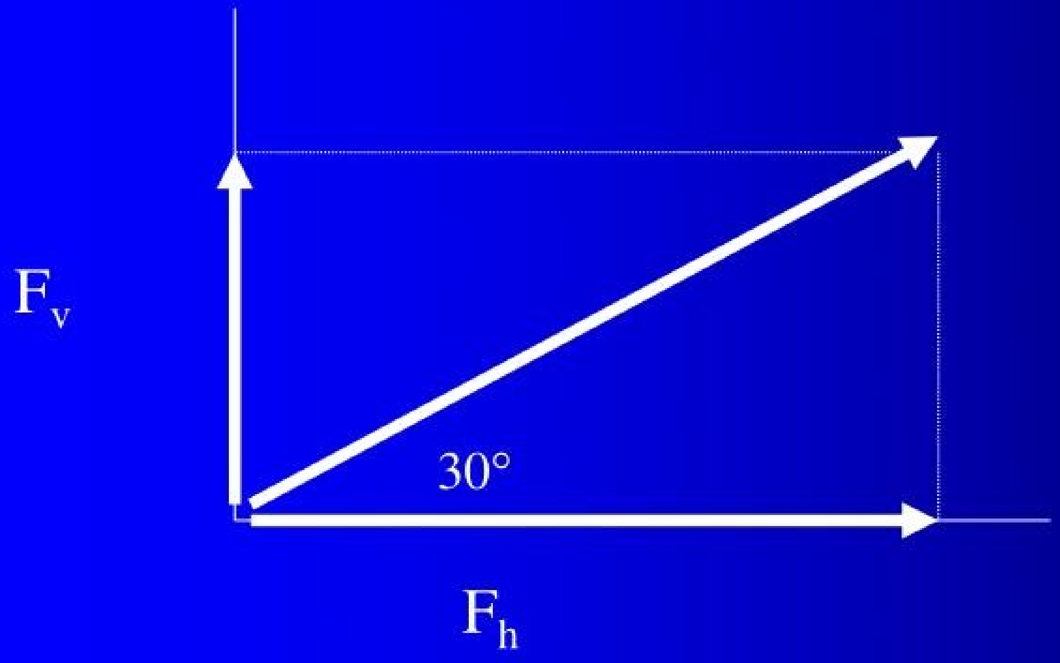
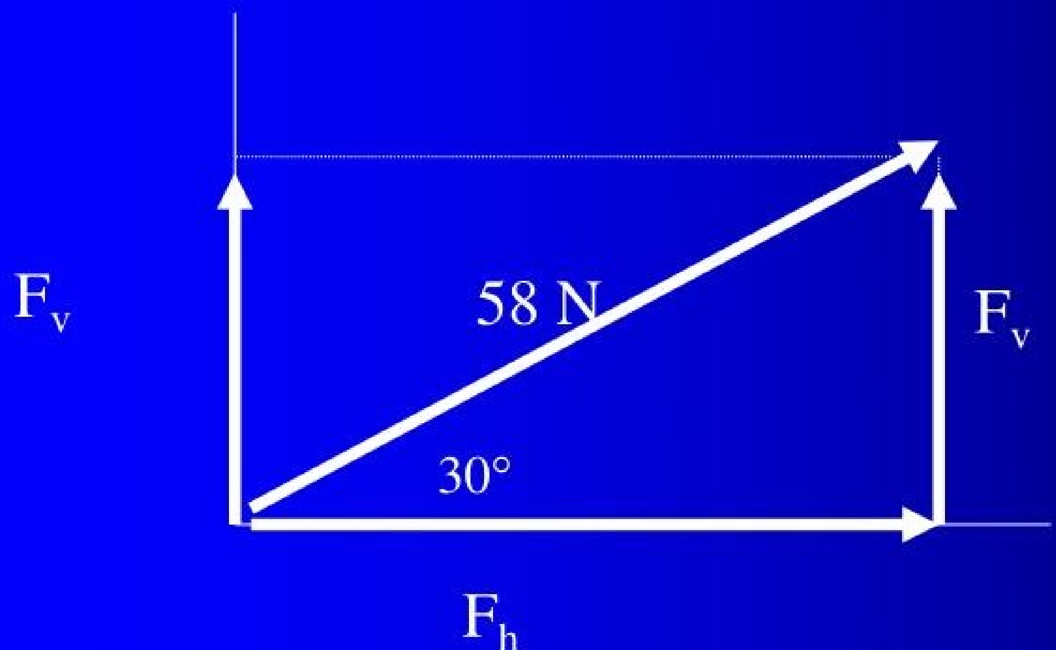


Vertical Component F_v

$\text{Sin } 30° = \frac{F_v}{58\text{ N}}$

$F_v = (\text{Sin } 30°)(58\text{ N})$

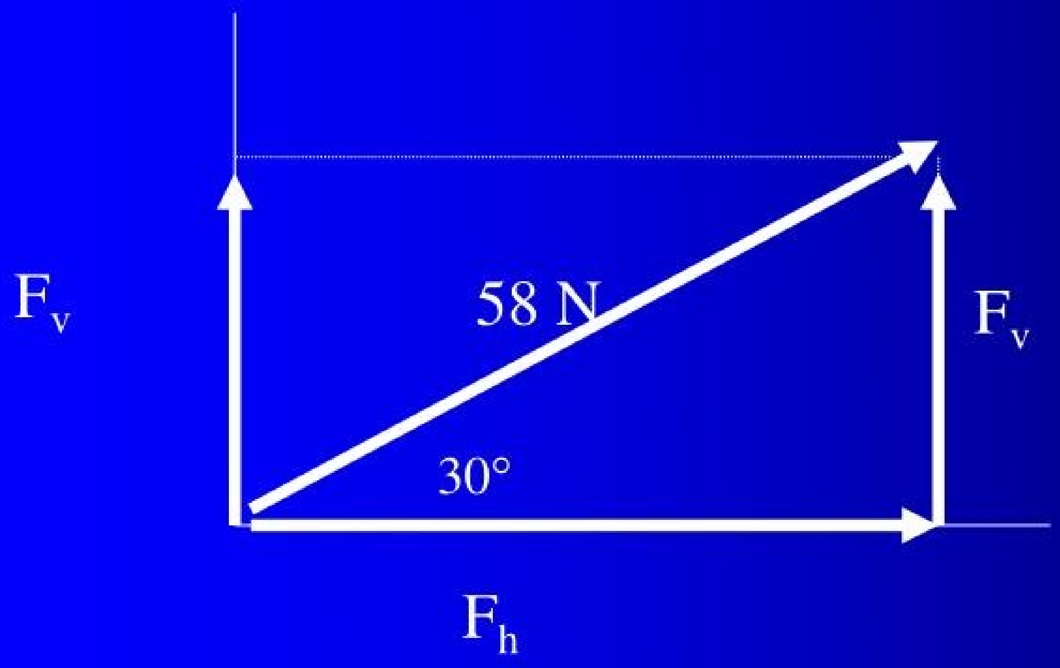
$F_v = 29\text{ N}$



Horizontal Component F_h

$$\cos 30^\circ = \frac{F_h}{58\ N} \qquad F_h = (\cos 30^\circ)(58\ N)$$

$$F_h = 50.2\ N$$



$F_h = 50.2$ N

$F_v = 29$ N